

X-ray imaging on pigments

Timepix4 project, Si sensor

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Introduction

The Timepix4 detector is employed to perform X-ray imaging on samples of different pigments on wood, aiming to identify their K-edges and characterise the instrumental response and the possible limitations of such analytical technique in a low energy range, hence using a silicon sensor. This characterisation is to be added to the previous one on different pigments at higher energies (employing a cadmium telluride sensor).

Experimental setup

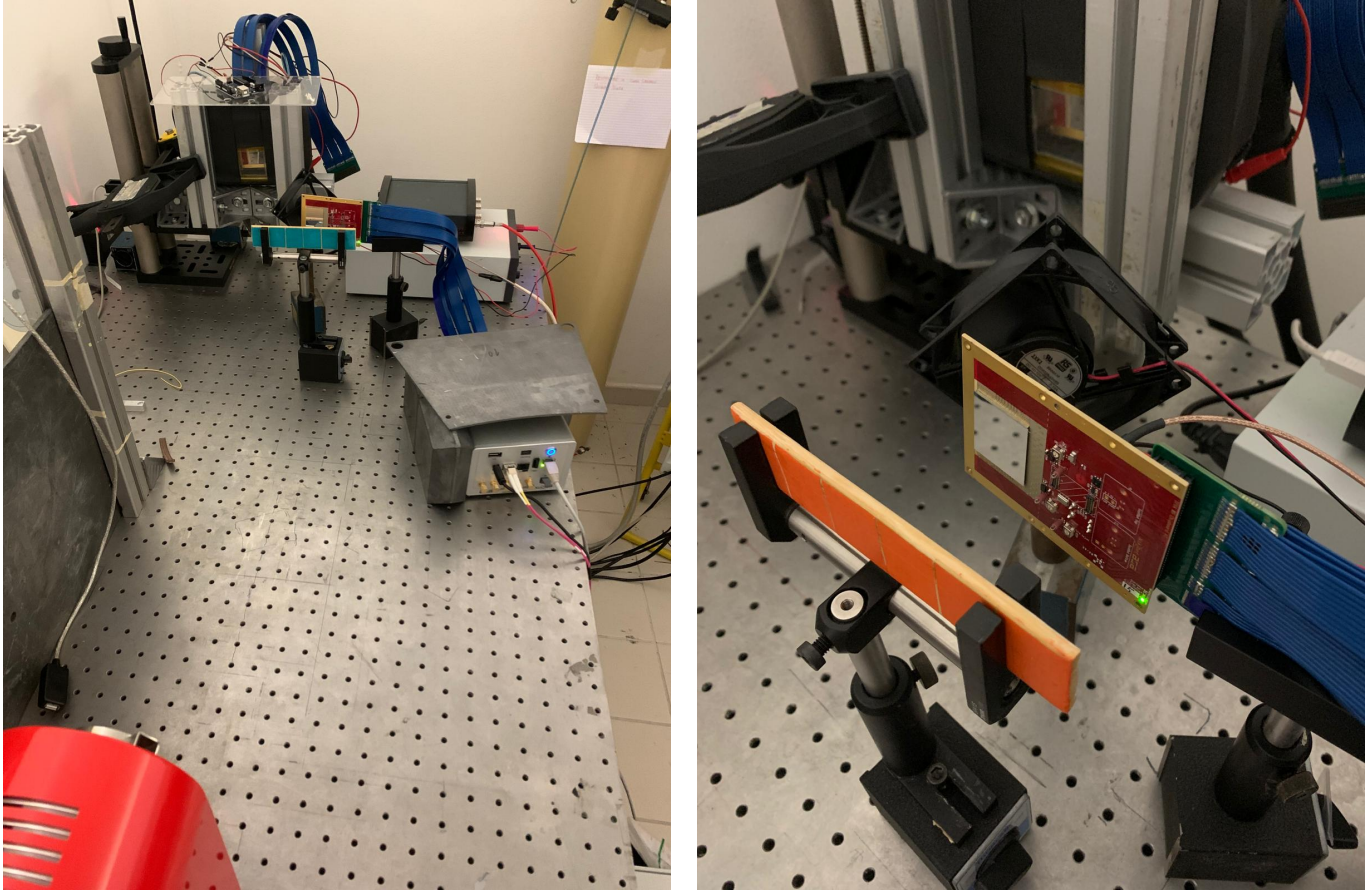


Figure 1: Experimental setup; from bottom to top: X-ray tube, sample, Timepix4 in electrical contact with the read-out system (shielded by lead plates); no cooling system is required for the silicon sensor

The X-ray detector is placed behind the sample, at an approximate distance of 7 *cm*, and the distance between the source and the detector is approximately 75 *cm*.

Source

The employed source is a micro-focus X-ray tube (Hamamatsu), characterised by:

- tungsten anode in transmission mode
- energy range: $kVp = 40 \text{ keV}$ for all measurements
- anode current: $I = 10 \mu A$
- acquisition time: 3 runs, each with an exposition of 100 s (total acquisition time of 300 s)
- no filters and no collimator are used

Sample

Samples are pigments and binders spread on wood tables of width $5 \pm 1 \text{ mm}$:

- cadmium-based pigments: Cadmium Red ($CdS \cdot CdSe$), Cadmium Orange ($CdS \cdot CdSe$), Dark Cadmium Red ($CdS \cdot CdSe + \text{unknown compound}$), Cadmium Green ($CdS + Al_2CoO_4$)
- arsenic-based pigment: Orpiment (As_2S_3)
- copper-based pigments: Azurite ($2CuCO_3 \cdot Cu(OH)_2$), Malachite ($CuCO_3 \cdot Cu(OH)_2$), copper-Phthalocyanine ($Cu(C_{32}H_{16}N_8)$)
- cobalt-based pigment: Cobalt Blue ($CoO \cdot Al_2O_3$)
- iron and manganese-based pigments: Terra d'Ombra ($Fe_2O_3 \cdot MnO_2 \cdot Al_2O_3 \cdot SiO_2$), Terra Verde di Verona ($[Fe, Mg](AlSi_3, Si_4)O_{10}(OH)_2$)
- chromium-based pigment: Guignet Green ($Cr_2O_3 \cdot (H_2O)_2$)
- binders on wood: egg, oil



Figure 2: Analysed pigments (on the right, two gradations of Cadmium Orange and two gradations of Cadmium Red)

Detection system

Timepix4 is used with a Si detector, so that the detection system is characterised by:

- Si thickness: $300\ \mu\text{m}$
- expected energy resolution: $2\ \text{keV} < FWHM < 5\ \text{keV}$
- pixel area: $55 \times 55\ \mu\text{m}^2$
- detector area: 448×512 pixels, i.e. total detector area of $6.94\ \text{cm}^2$
- readout system: SPDR4 (Speedy Pixel Detector Readout)

Methods

The X-ray spectrum is firstly acquired without the sample (flat field acquisition with $kVp = 40\ \text{keV}$), although the ultimately employed normalising attenuation signals are binders on wood; in any case, the pigment sample is then inserted and the I spectrum is acquired. The applied calibration parameters were derived using a combination of fluorescence photons (from higher to lower energies, i.e. positions of fluorescence peaks of tin, molybdenum, zirconium, zinc, copper, cobalt, iron) and test pulses (calibration with electric signals).

A Timepix4 acquisition in data-driven mode allows to obtain the position (x, y) of each occurring event, the Time of Arrival and the Time over Threshold (ToA, ToT) of events with a fixed threshold of $700\ e^-$. Information gained through position, ToT and ToA is combined to perform clustering of detected events, i.e. to group hits in clusters on pixels, and to measure the energy of incident radiation. For the purposes of this work, energy information on counts and clustering of events are only taken into consideration.

For each 100 s-lasting run, acquired energy information on counts is clustered and processed to obtain nine different sets of 40 subsequent stacked images (TIFF format), corresponding to energy slicing in the range $[0\ \text{keV}, 40\ \text{keV}]$ with an energy bin width of $1\ \text{keV}$. Each different set of stacks is associated to a fixed cluster size: $\{1, 2, 3, 4, 5, 6, 7, 8\}$ and “all” (i.e. superposition of signals from all cluster sizes). To avoid undesired effects due to the time variation of clustered events percentages, only signals with cluster size “all” are taken into account.

For each acquired pigment on wood transmission signal, the respective I_0 file is compared to the I file, so to derive total attenuation spectra and extrapolate information about the expected K-edges. Therefore, for cluster size “all”, stacks are processed in a final $-\ln(I/I_0)$ spectrum.

To enhance K-edge visibility, subtraction analyses (K-edge subtraction or differential imaging) are performed on each pigment’s total attenuation clustered data. An energy bin width of $1\ \text{keV}$ is kept throughout the analyses.

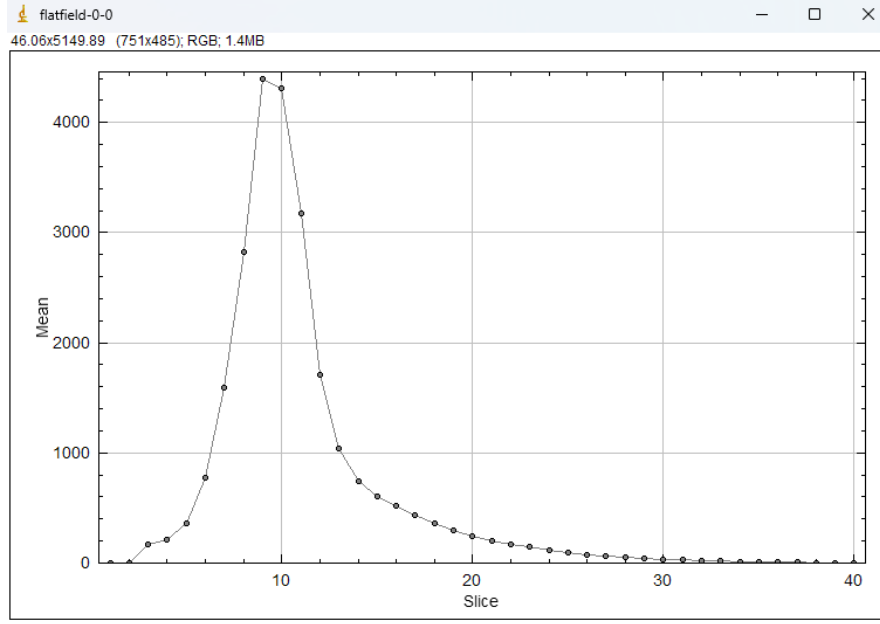
I_0 signal generation

In order to minimise artefacts and noise in pigments’ attenuation signals caused by the non-uniformity of wood, the actually employed I_0 stacks are extracted from the average intensity values (over the whole slice area) of egg and oil on wood. In particular, a custom `Python` script is used to generate spatially homogeneous slices in the energy range $[0\ \text{keV}, 40\ \text{keV}]$, representing the mean counts on a binder on wood ROI (excluding flat field counts). A normalising stack is generated for both egg and oil on wood; total attenuation imaging results obtained with mean counts of egg on wood, mean counts of oil on wood, and flat field as I_0 are compared.

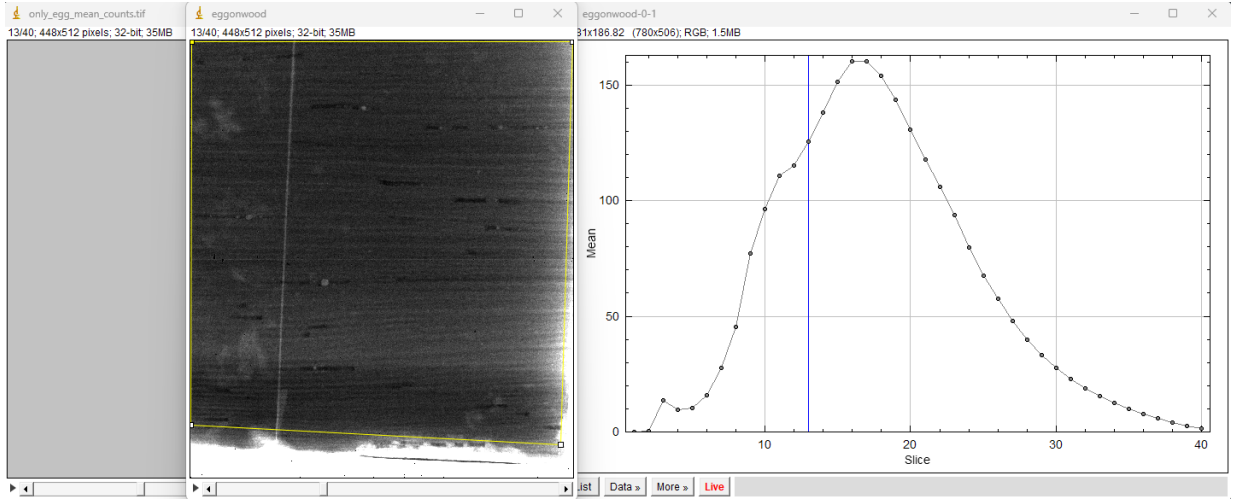
Results

Results are reported for each analysed pigment, grouping them by their characteristic chemical element. Total attenuation signals obtained using flat field as I_0 are not reported, as they usually do not allow to clearly identify K-edges, because of the wood contribution to the sample attenuation. Results of $-\ln(I/I_0)$ being I_0 the egg on wood signal or the oil on wood signal are overall similar (if not undistinguishable), hence attenuation results with these two possible normalising signals are reported indifferently.

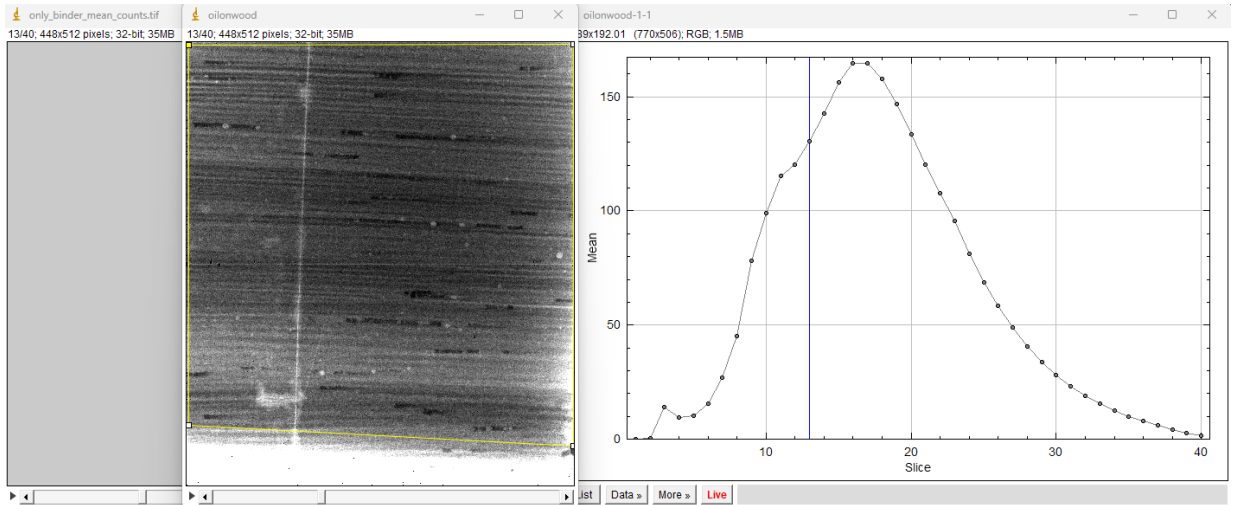
Flat field and binders



(a) Flat field counts, 40 kVp



(b) Egg on wood counts, 40 kVp ; on the left, the generated spatially homogeneous I_0 stack (mean counts of the selected original ROI, appearing as a sequence of grey slices)



(c) Oil on wood counts, 40 kVp ; on the left, the generated spatially homogeneous I_0 stack (mean counts of the selected original ROI, appearing as a sequence of grey slices)

Figure 3: Flat field and binders counts, to be used as I_0 signals

Cadmium-based pigments

As it can be seen in the figures below, the presence of cadmium can be identified in all four pigments. In particular, cadmium's K-edge seems to be quite evident even from total attenuation signals, but Kes analyses are extremely relevant to confirm the identification of the expected K-edge, as they all show a peak at slice 27, which finely matches the theoretical K-edge at $E_K = 26.711 \text{ keV}$. The lower intensity of Kes peaks in Dark Cadmium Red and Cadmium Green is consistent with the fact that a lower concentration of cadmium sulphide is expected in these pigments.

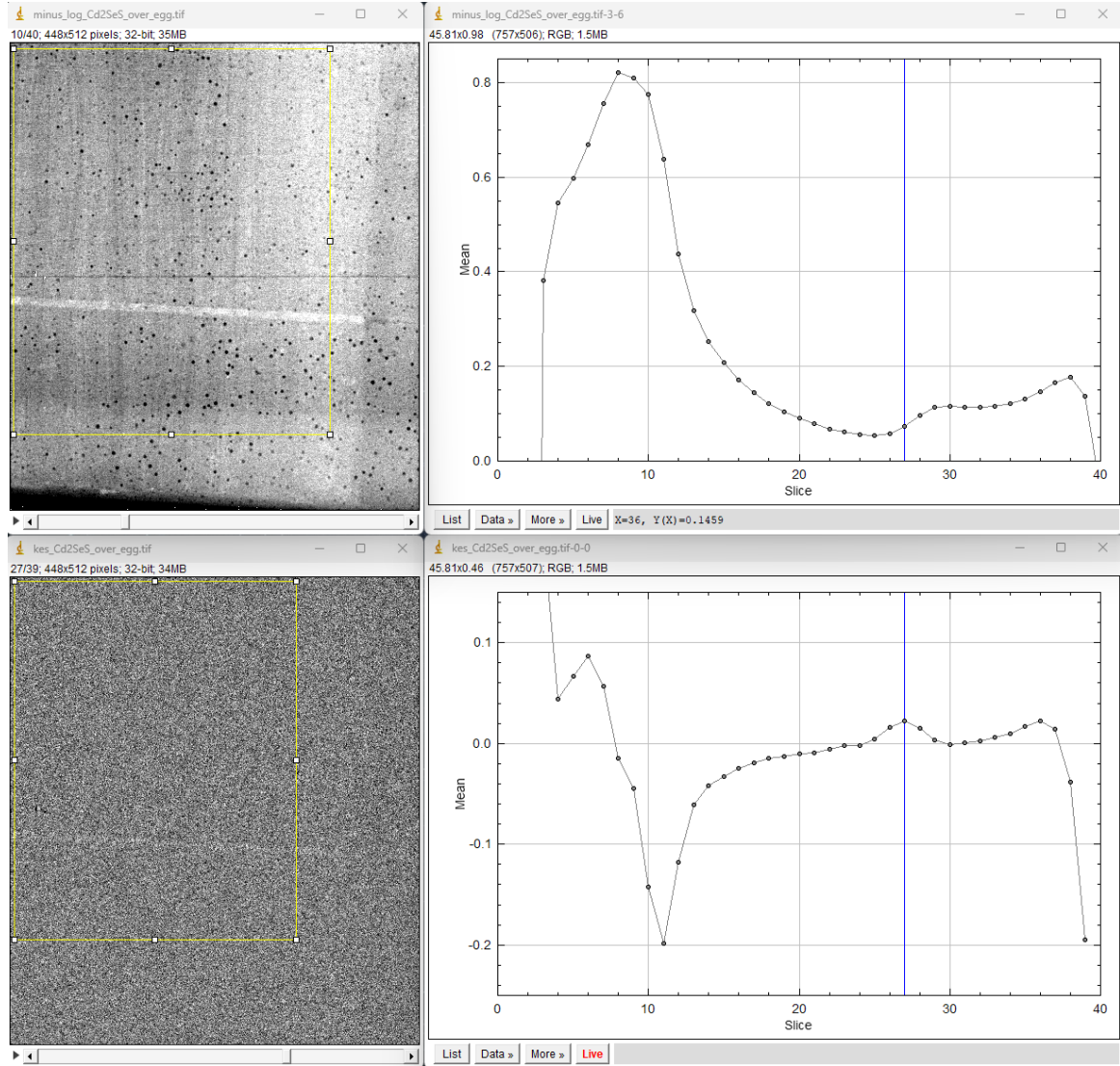
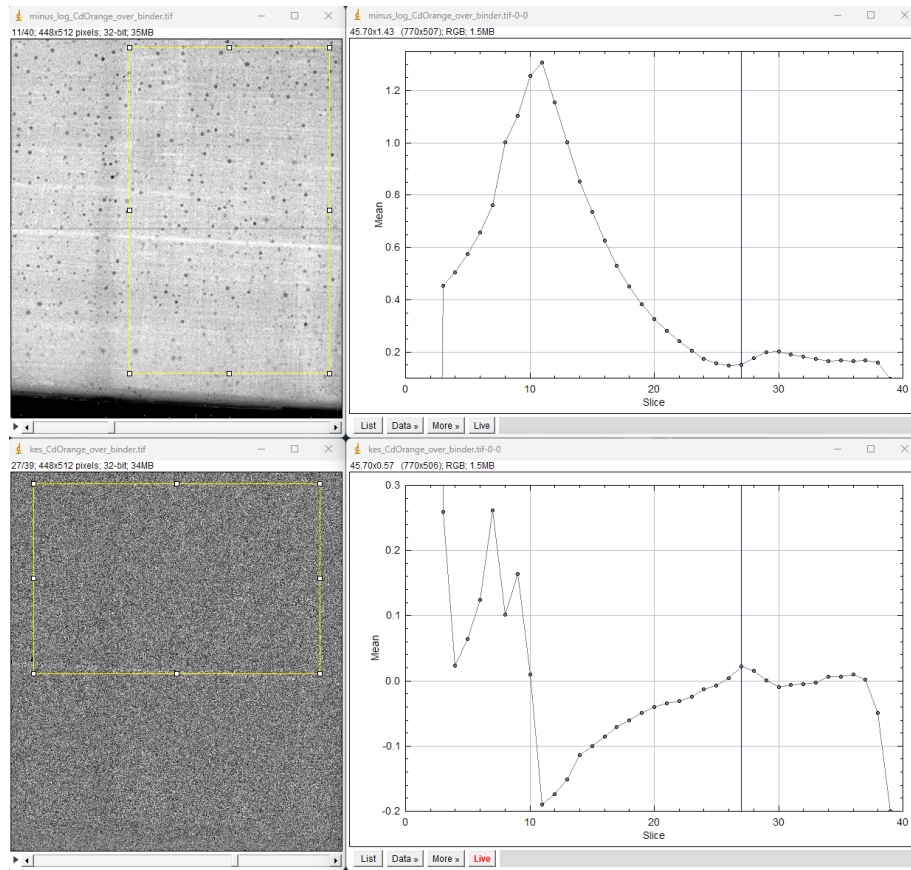
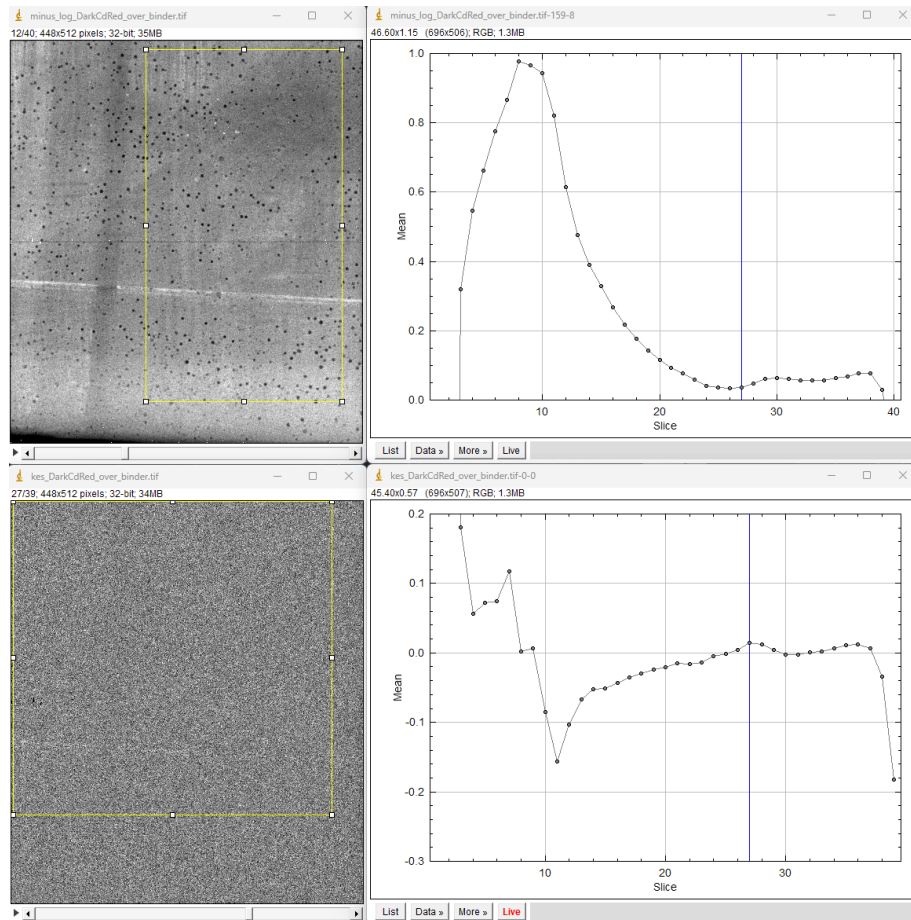


Figure 4: Cadmium Red total attenuation and K-edge subtraction, 40 kVp



(a) Cadmium Orange total attenuation and K-edge subtraction, 40 kVp



(b) Dark Cadmium Red total attenuation and K-edge subtraction, 40 kVp

Figure 5

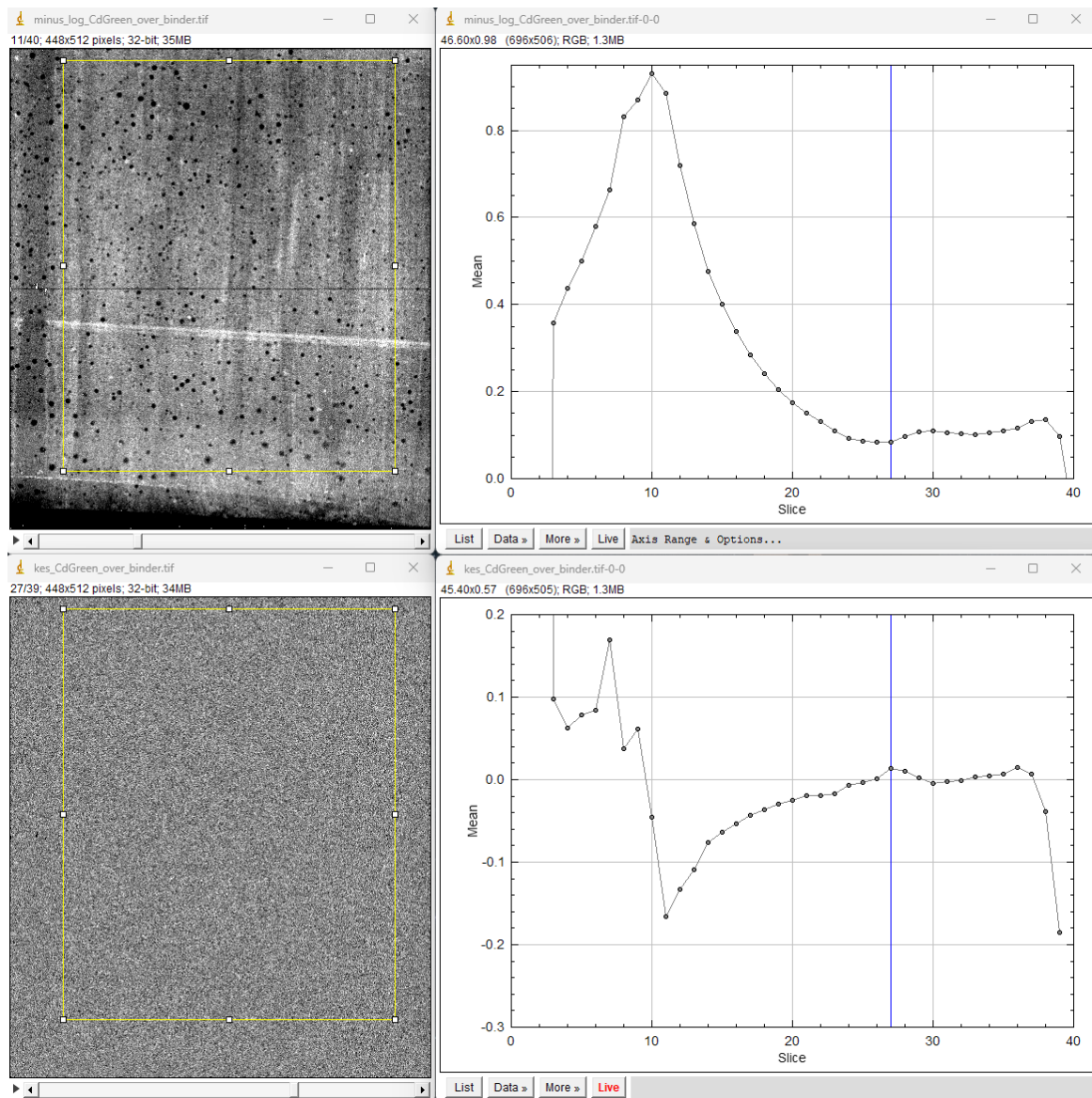


Figure 6: Cadmium Green total attenuation and K-edge subtraction, 40 kVp

Arsenic and selenium-based pigments

As it can be noted in Figure 6, the presence of arsenic in Orpiment is not evident from the total attenuation plot (despite the change in the slope from slice 11 to slice 14), but such element can be identified with K-edge subtraction analysis: the observed peak at slice 12 is compatible with the expected arsenic K-edge at $E_K = 11.867 \text{ keV}$.

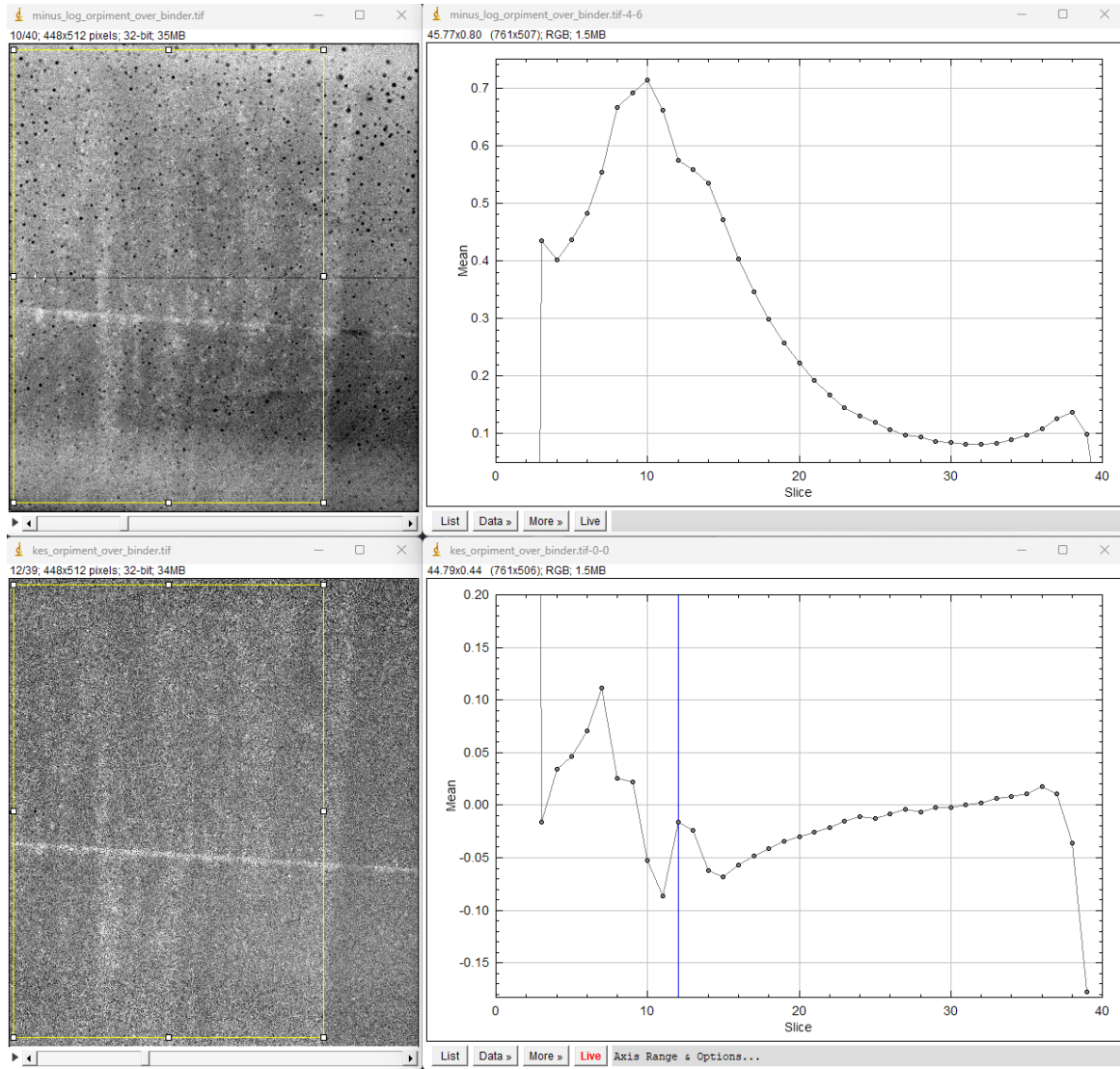
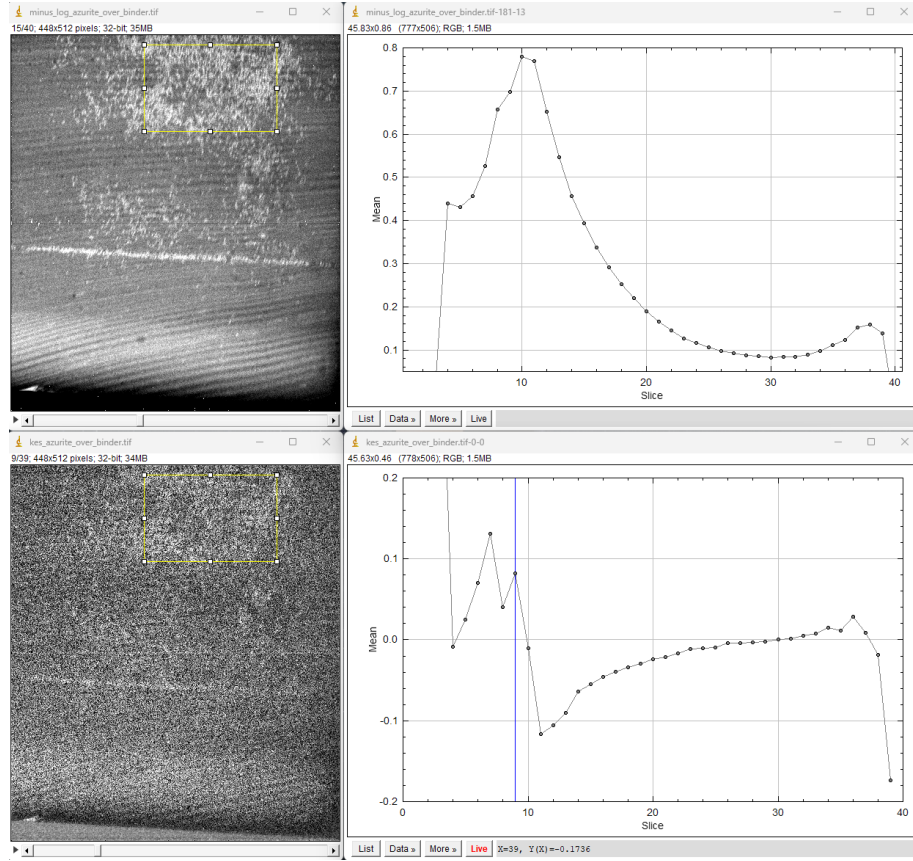


Figure 7: Orpiment total attenuation and K-edge subtraction, 40 kVp

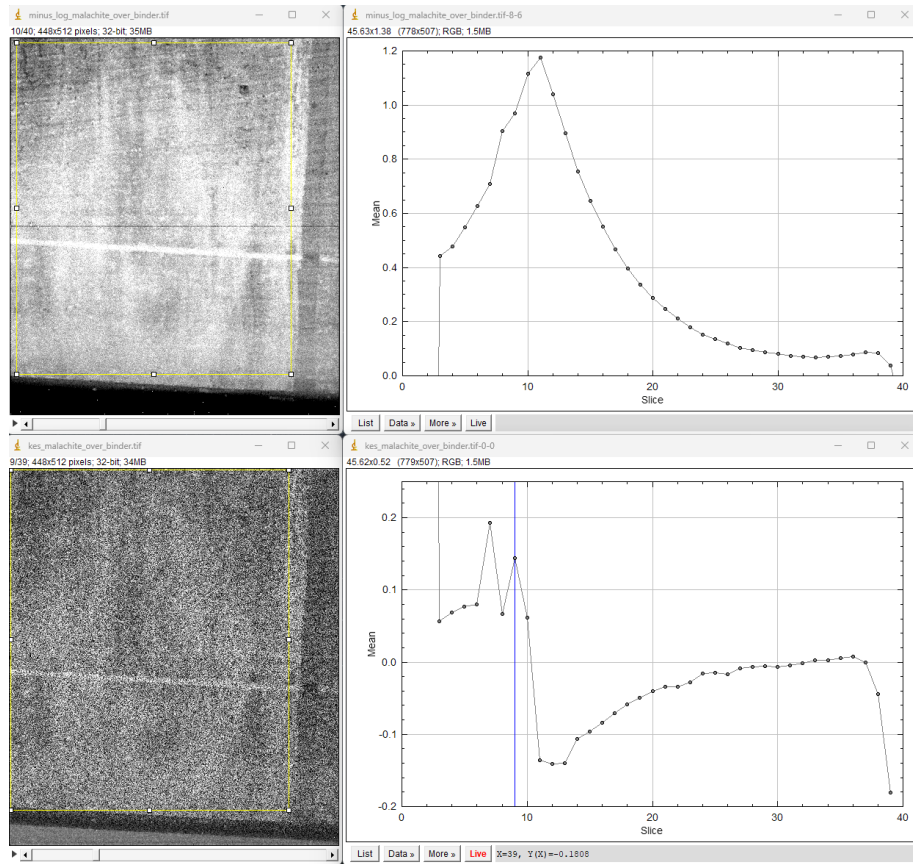
In principle, with the selected energy range the presence of selenium in Cadmium Red, Cadmium Orange and Dark Cadmium Red could be detected (selenium's K-edge is expected at $E_K = 12.658 \text{ keV}$). In particular, the concentration of cadmium selenide is typically smaller in orange cadmium pigments and higher in red cadmium pigments. Nevertheless, selenium is not identifiable in any of these pigments, as it can be seen in Figure 4 and in Figure 5. Since arsenic, whose K-edge is close to the selenium one, could be clearly spotted, the impossibility of detecting selenium in cadmium-based pigments might be due to its concentration being too small to produce a significant effect in total attenuation (or, in general, to the lower concentration of selenium with respect to cadmium in the analysed pigments).

Copper-based pigments

Copper K-edge is expected at $E_K = 8.979 \text{ keV}$: in all analysed pigments (Azurite, Malachite, copper-Phthalocyanine) a peak is displayed at slice 9 of Kes images, which might confirm the presence of copper, otherwise undetectable when analysing just the total attenuation spectra. It should also be taken into account, though, that at low energies Kes peaks can be unreliable. As a matter of fact, the peak at slice 9 could be due to noise, as well as the peak at slice 7: no elements with K-edges expected around 7 keV are supposed to be contained in the considered pigments.



(a) (traces of) Azurite total attenuation and K-edge subtraction, 40 kVp



(b) Malachite total attenuation and K-edge subtraction, 40 kVp

Figure 8

The similar appearance of total attenuation spectra (using oil on wood as the I_0 signal) of Azurite and Malachite is consistent with the similar densities and chemical properties characteristic of these two pigments.

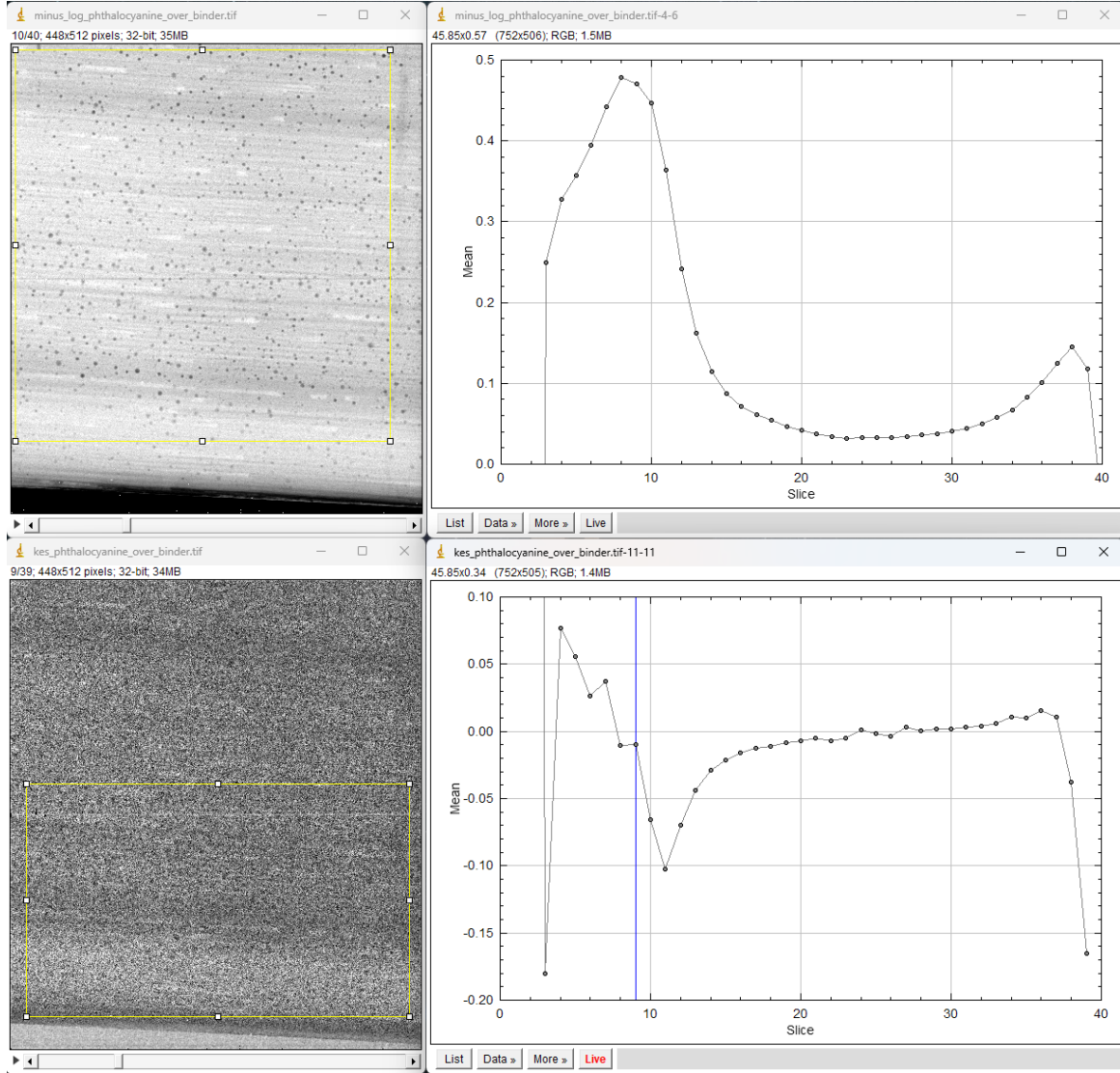
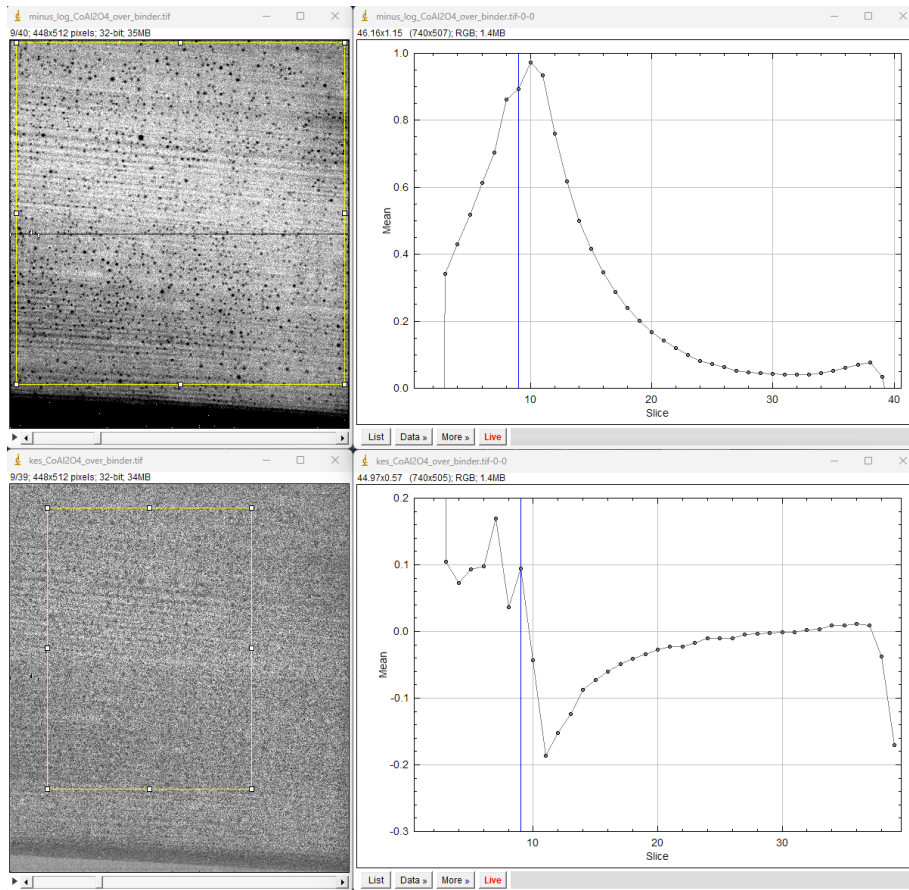


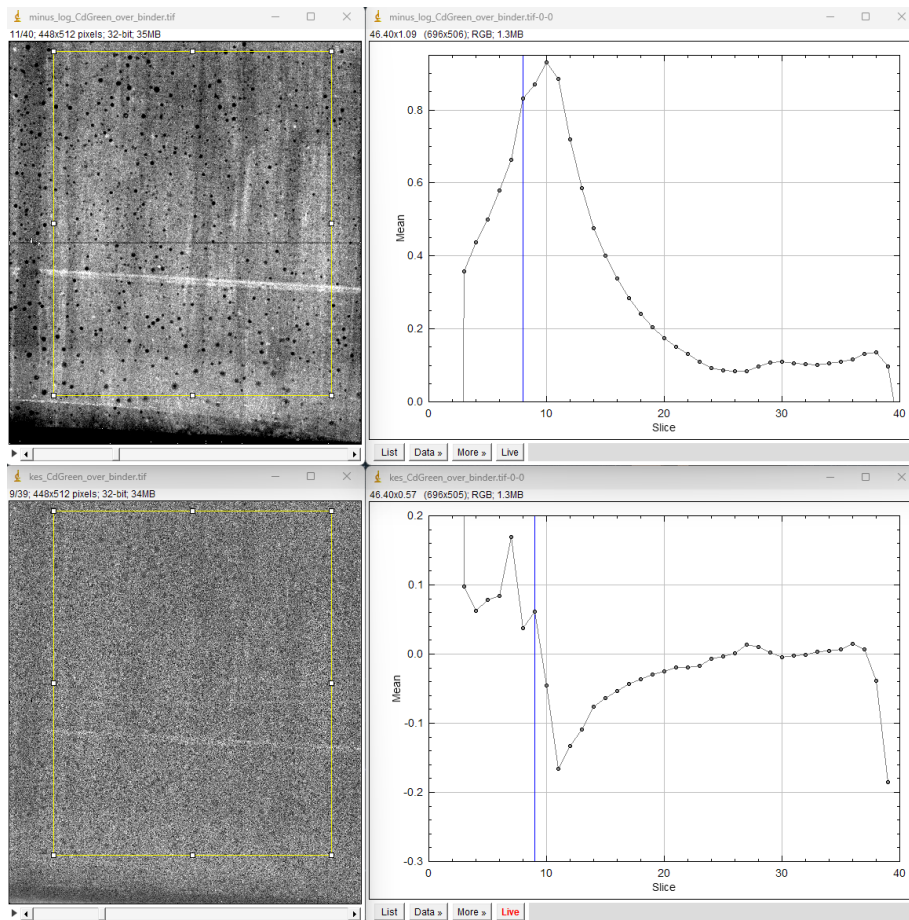
Figure 9: Phthalocyanine total attenuation and K-edge subtraction, 40 kVp

Cobalt-based pigments

Cobalt K-edge is expected at $E_K = 7.709 \text{ keV}$: no unambiguous increasing profile in total attenuation signals nor peak around 8 keV in Kes images can be seen, hence cobalt might be out of the detection range of this setup. However, it is not to be excluded that, due to the limited energy resolution, a possible cobalt peak in Kes imaging is displayed at slice 9. In particular, resolution is quantified with the previously measured fluorescence peaks of copper and zinc (i.e. approximately in the range $[8 \text{ keV}, 10 \text{ keV}]$), and it turns out to be characterised by $FWHM \simeq 2.3 \text{ keV}$. Despite this hypothesis, there is still a good chance peaks at slice 9 (as well as slice 7) are noisy artefacts.



(a) Cobalt Blue total attenuation and K-edge subtraction, 40 kVp



(b) Cadmium Green total attenuation and K-edge subtraction, 40 kVp

Figure 10

Iron and chromium-based pigments

In this case, expected K-edges are: $E_K = 7.112 \text{ keV}$ for iron, $E_K = 6.539 \text{ keV}$ for manganese and $E_K = 5.989 \text{ keV}$ for chromium, and because of the energy resolution in this low energy range ($FWHM \simeq 3.5 \text{ keV}$) these three elements could in principle be undistinguishable. There seems to be a peak in Kes images of all analysed pigments at slice 7, which might be compatible with the position of iron, manganese and chromium K-edges. However, the appearance of a second unexpected peak at slice 9 in Terra Verde di Verona and the presence of the same peak at slice 7 in many other pigments (which are not supposed to be made of compounds with such a low K-edge) lead to think that signals observed at low energies are noisy artefacts, hence that the considered chemical elements cannot be reliably detected.

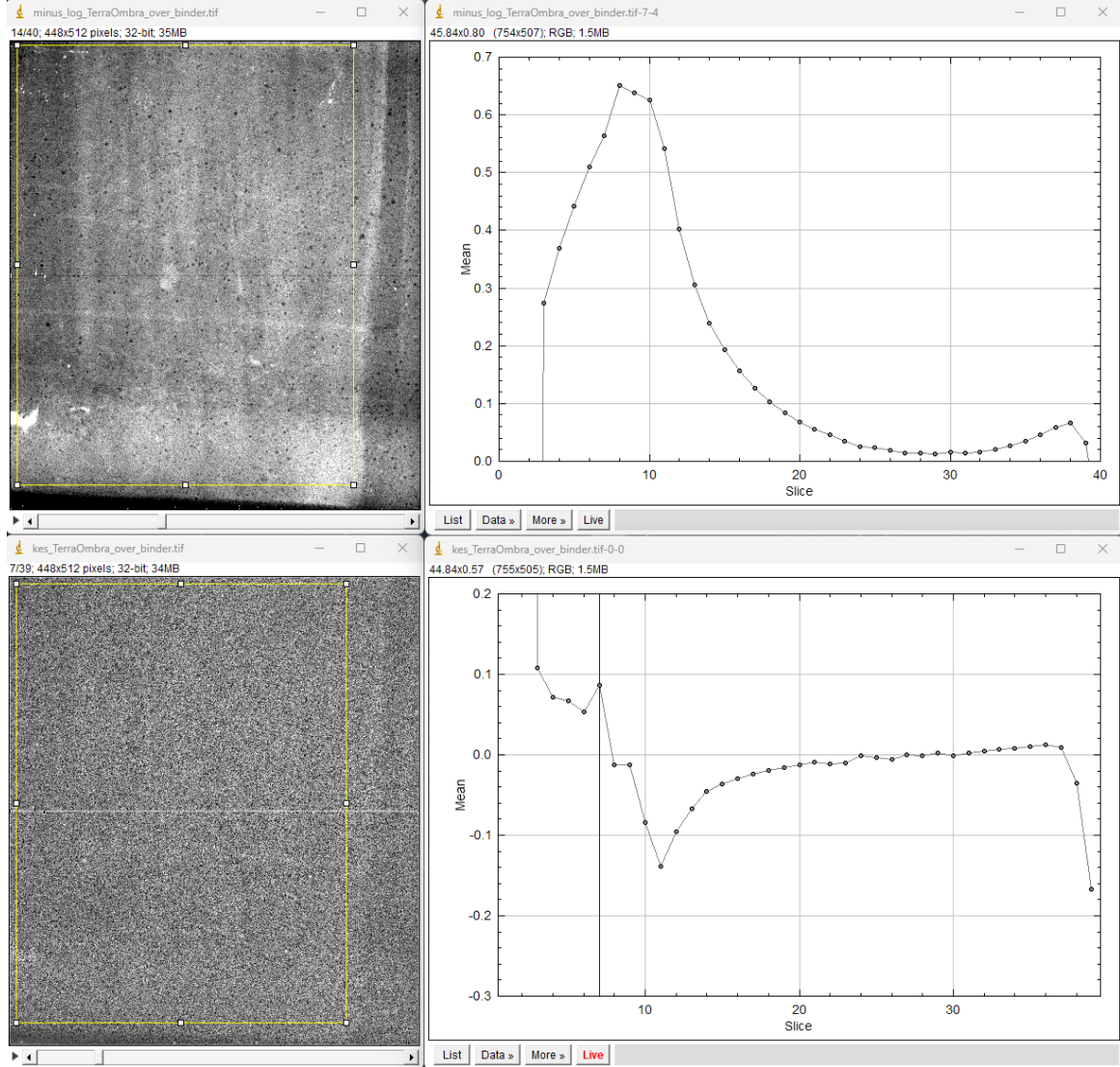
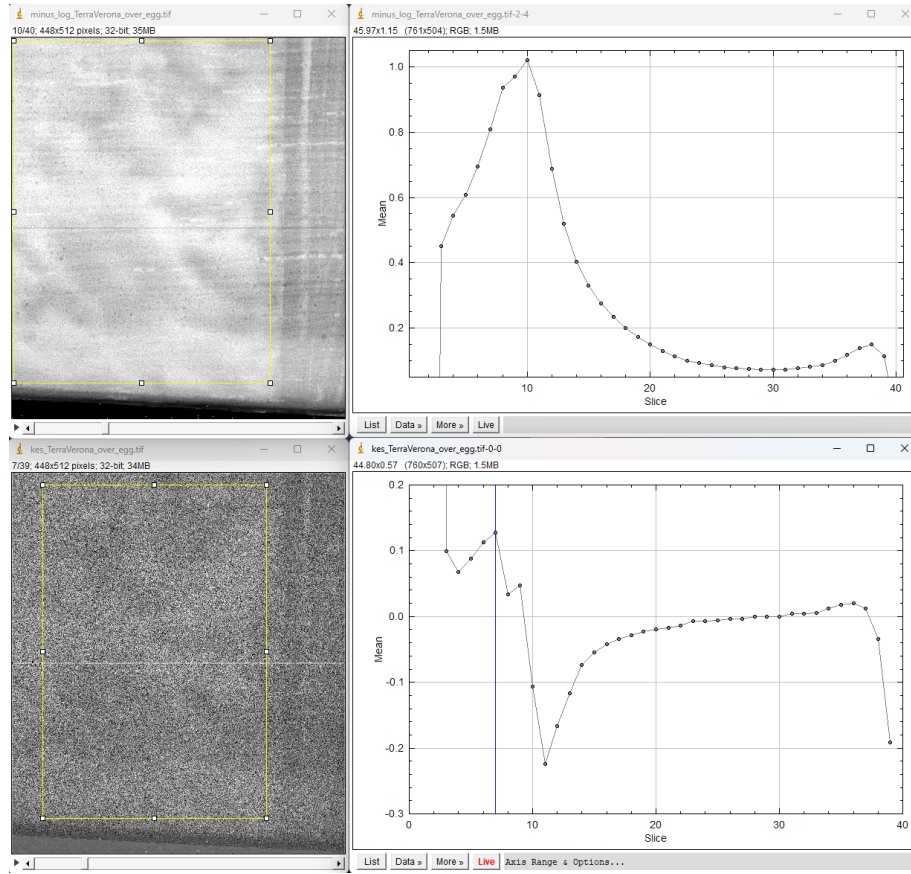
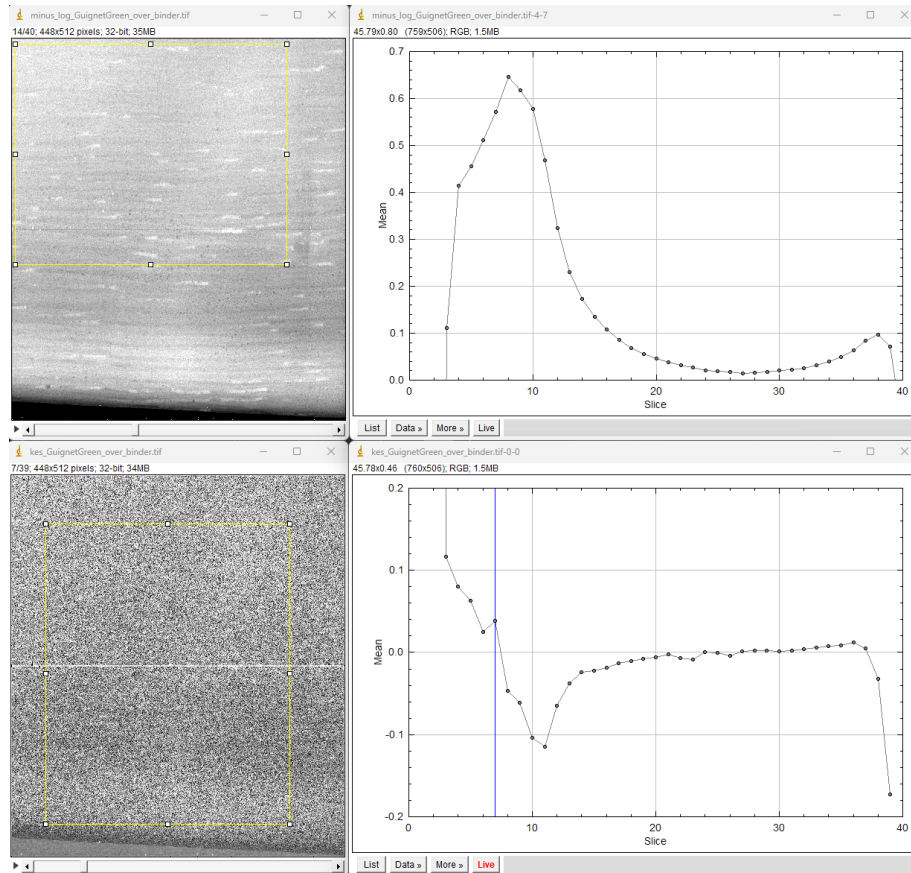


Figure 11: “Terra d’Ombra” total attenuation and K-edge subtraction, 40 kVp



(a) “Terra Verde di Verona” total attenuation and K-edge subtraction, 40 kVp



(b) Guignet Green total attenuation and K-edge subtraction, 40 kVp

Figure 12

Anomalies

Assuming that the energy resolution is not sufficient to distinguish between K-edges of copper and cobalt and between K-edges of iron, manganese and chromium, there are still some effects observed in Kes imaging that might lead to the conclusion that low-energy K-edges are not detectable, not even with a silicon sensor.

In particular, the peak at slice 7 appears in Kes images of each one of the twelve analysed pigments, but it was only expected in a minority of them (Terra d'Ombra, Terra Verde di Verona, Guignet Green, possibly Dark Cadmium Red, Cadmium Green, Cobalt Blue). It can then be deduced that the observed effect is systematic (possibly caused by low-energy noise) and therefore the employed analytical technique is not suitable to detect elements with low K-edges (or more in general light elements).

In addition to that, the peak at slice 9 appears not only in Kes images of the expected pigments (Azurite, Malachite, Phthalocyanine, possibly Cadmium Green and Cobalt Blue) but also in Kes images of Cadmium Orange and Terra Verde di Verona: although it cannot be treated as a systematic effect, the possibility of it being originated from noise should not be excluded. In the worst case, the employed technique is not suitable to detect elements lighter than zinc.

Thickness estimation

In principle, thickness of pigments could be estimated from total attenuation results, starting from measurements of the $-\ln(I/I_0)$ signals at fixed energies and from tabulated attenuation coefficients of known elements. Thicknesses of all analysed pigments have been estimated (using either egg or oil on wood as the I_0 signal) in correspondence of K-edges when possible (i.e. for cadmium-based pigments and Orpiment) and in the range $[10\text{ keV}, 15\text{ keV}]$ for copper, cobalt, iron and chromium-based pigments (since their K-edges are not undeniably recognisable and tungsten fluorescence photons at low energies might alter the acquired transmission signals). All pigments seem to result in being roughly $[1\text{ }\mu\text{m}, 10\text{ }\mu\text{m}]$ thick, but such estimations are not expected to be accurate: wood thickness estimation using the same method yields $d_{\text{wood}} \sim 2.7\text{ cm}$, but the directly measured wood width is $d_{\text{wood}} = 0.5 \pm 0.1\text{ mm}$. Although the rough assumption of an average wood attenuation coefficient definitely plays a role in the latter result, estimated pigments' thicknesses do not seem to be consistent with results obtained starting from measurements with the CdTe sensor (in particular, pigments were expected to be thicker than what found out).

Conclusions

Putting all results together, it is evident that pigments characterisation with Timepix4 in the described experimental configuration can be considered to be accurate only if the energy range of interest is not below approximately 11 keV . As a matter of fact, the element with the lowest K-edge that could be detected up to a sufficient precision is arsenic ($E_K \simeq 11.9\text{ keV}$). There is a chance characterisation could extend down to copper, but further analyses (possibly modifying anodic current, filtration and time of acquisitions) are required to deepen such hypothesis.

Links

- [1] Timepix4 specifications [here](#)
- [2] Tables of linear attenuation coefficients and material densities from NIST [here](#)
- [3] Material K-edges can be found in the X-ray data booklet [here](#)
- [4] Details about previous analyses [here](#)